DA-AG-018: Anomaly Detection & Time Series Assignment

Question 1: What is Anomaly Detection? Explain its types with examples.

Anomaly Detection is a data analysis technique used to identify unusual patterns or observations that do not conform to expected behavior. These unusual data points are called anomalies or outliers. Anomaly detection is widely used in fraud detection, network intrusion detection, fault detection in machines, healthcare monitoring, and financial analysis.

Types of anomalies:

1. Point Anomaly:
A single observation that significantly differs from the rest of the data.
Example: If the normal temperature of a machine ranges between 50°C and 60°C, but one reading shows 120°C, it is a point anomaly.

2. Contextual Anomaly:
An observation that is anomalous in a specific context such as time or location.
Example: High electricity consumption at 3 AM is abnormal, but the same consumption at 3 PM may be normal.

3. Collective Anomaly:
A group of observations that together form an anomaly, even if individual points are normal.
Example: Multiple failed login attempts within a short time period may indicate a cyber attack.

Question 2: Compare Isolation Forest, DBSCAN, and Local Outlier Factor.

Isolation Forest:
Approach: Tree-based model that isolates anomalies using random splits.
Use cases: Fraud detection, network security, large datasets.
Advantage: Fast and efficient for large and high-dimensional data.

DBSCAN:
Approach: Density-based clustering algorithm that identifies low-density points as anomalies.
Use cases: Spatial data analysis, clustering applications.
Advantage: Can detect arbitrary-shaped clusters.

Local Outlier Factor (LOF):
Approach: Compares local density of a point with its neighbors.
Use cases: Detecting local anomalies in datasets with varying density.
Advantage: Effective in identifying local outliers.

Question 3: Key components of Time Series.

Trend: Long-term increase or decrease in data values.
Example: Increase in airline passengers over years.

Seasonality: Regular repeating patterns at fixed intervals.
Example: Higher sales during festive seasons every year.

Cyclic: Long-term oscillations due to economic cycles.
Example: Business profits increasing and decreasing over years.

Residual: Random noise remaining after removing trend and seasonality.
Example: Unexpected spikes due to random events.

Question 4: Stationarity in Time Series.

A stationary time series is one whose statistical properties such as mean, variance, and autocorrelation remain constant over time.

Testing methods:
- Augmented Dickey-Fuller (ADF) test
- KPSS test

Transformation methods:
- Differencing
- Log transformation
- Removing trend and seasonality

Question 5: Difference between AR, MA, ARIMA, SARIMA, SARIMAX.

AR: Uses past values to predict future values.
MA: Uses past prediction errors.
ARIMA: Combines AR, MA, and differencing.
SARIMA: Includes seasonal components.
SARIMAX: Includes seasonal and external variables like weather.

Question 6: Load and decompose AirPassengers dataset.

The dataset is plotted and decomposed into trend, seasonality, and residual components.

Python Code:

```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose

data = pd.read_csv('AirPassengers.csv', parse_dates=['Month'],
index_col='Month')

data.plot(title='Original Time Series')
plt.show()

result = seasonal_decompose(data['Passengers'], model='multiplicative',
period=12)
result.plot()
plt.show()
```

Question 7: Isolation Forest anomaly detection.

Isolation Forest identifies anomalies by isolating rare data points.

Python Code:

```
from sklearn.ensemble import IsolationForest
import matplotlib.pyplot as plt

model = IsolationForest(contamination=0.05, random_state=42)
data['anomaly'] = model.fit_predict(data[['fare_amount']])

plt.scatter(data.index, data['fare_amount'], c=data['anomaly'])
plt.title('Isolation Forest Anomalies')
plt.show()
```

Question 8: SARIMA Forecast.

SARIMA model is trained and used to forecast next 12 months.

Python Code:

```
from statsmodels.tsa.statespace.sarimax import SARIMAX

model = SARIMAX(data['Passengers'], order=(1,1,1),
seasonal_order=(1,1,1,12))
result = model.fit()

forecast = result.forecast(steps=12)

forecast.plot(title='12 Month Forecast')
plt.show()
```

Question 9: LOF anomaly detection.

LOF detects anomalies based on local density differences.

Python Code:

```
from sklearn.neighbors import LocalOutlierFactor

lof = LocalOutlierFactor(n_neighbors=20)
data['anomaly'] = lof.fit_predict(data[['Passengers']])

plt.scatter(data.index, data['Passengers'], c=data['anomaly'])
plt.title('LOF Anomaly Detection')
plt.show()
```

Question 10: Real-time workflow.

Anomaly detection can be done using Isolation Forest. SARIMAX is best for forecasting because it uses external features.

Validation:
- Rolling validation
- RMSE and MAE monitoring

Business benefits:
- Prevent failures
- Improve efficiency
- Reduce cost

Python Code:

```
from sklearn.ensemble import IsolationForest
from statsmodels.tsa.statespace.sarimax import SARIMAX

anomaly_model = IsolationForest()
anomalies = anomaly_model.fit_predict(energy_usage.values.reshape(-
1,1))

model = SARIMAX(energy_usage, exog=weather, order=(1,1,1),
seasonal_order=(1,1,1,96))
forecast = model.fit().forecast(steps=96)
```